3.ipynb

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Code

•[1]:

```python
import time

class BankAccount:

    def __init__(self, card_number, pin, balance, owner):
        self.card_number = card_number
        self.__pin = pin
        self.balance = balance
        self.owner = owner
        self.is_locked = False

    def verify_pin(self, entered_pin):

        if self.is_locked:
            print("Card is locked. Contact bank support.")
            return False
        return entered_pin == self.__pin

    def update_balance(self, amount):

        self.balance += amount

    def get_balance(self):

        return self.balance

    def lock_card(self):

        self.is_locked = True

class ATM:


    ACCOUNT = BankAccount(
        card_number="1234567890123456",
        pin="4321",
        balance=7000000,
        owner="Sarmistha"
    )

    MAX_ATTEMPTS = 3

    def __init__(self):
        self.current_account = None
        self.pin_attempts = 0

    def insert_card(self, card_number):

        if card_number == self.ACCOUNT.card_number:
```

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Code

```python
def insert_card(self, card_number):

    if card_number == self.ACCOUNT.card_number:
        self.current_account = self.ACCOUNT
        self.pin_attempts = 0
        print(f"Card inserted. Welcome, {self.current_account.owner}!")
        return True
    else:
        print("Card not recognized.")
        return False

def authenticate(self):

    if not self.current_account:
        return False

    while self.pin_attempts < self.MAX_ATTEMPTS:
        pin_input = input("Enter your 4-digit PIN: ")

        if self.current_account.verify_pin(pin_input):
            print(" Authentication successful.")
            return True
        else:
            self.pin_attempts += 1
            attempts_left = self.MAX_ATTEMPTS - self.pin_attempts
            if attempts_left > 0:
                print(f"Invalid PIN. {attempts_left} attempts remaining.")


    if not self.current_account.is_locked:
        self.current_account.lock_card()
    print("Too many failed attempts. Card retained/locked.")
    self.current_account = None
    return False

def check_balance(self):
    """Displays the current account balance."""
    if self.current_account:
        print(f"\ Your current balance is: 700000{self.current_account.get_balance():.2f}")
    else:
        print("Please insert card and authenticate first.")

def withdraw(self, amount):

    if not self.current_account:
        print("Please insert card and authenticate first.")
        return

    if amount <= 0:
        print("Withdrawal amount must be positive.")
```

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Code

```
        if not self.current_account:
            print("Please insert card and authenticate first.")
            return

        if amount <= 0:
            print("Withdrawal amount must be positive.")
        elif amount % 10 != 0:
            print("Please enter an amount divisible by 10.")
        elif amount > self.current_account.get_balance():
            print("Insufficient funds. Transaction cancelled.")
        else:

            self.current_account.update_balance(-amount)
            print(f" Dispensing ${amount:.2f}...")
            time.sleep(1)
            print("Please take your cash.")
            self.check_balance()

    def run_session(self):

        atm = ATM()


        if not atm.insert_card(self.ACCOUNT.card_number):
            return


        if not atm.authenticate():
            return


        while True:
            print("\n--- ATM Menu ---")
            print("1. Check Balance")
            print("2. Withdraw Cash")
            print("3. Exit/Eject Card")

            choice = input("Enter choice (1/2/3): ")

            if choice == '1':
                atm.check_balance()
            elif choice == '2':
                try:
                    withdrawal_amount = int(input("Enter amount to withdraw (multiples of 10): "))
                    atm.withdraw(withdrawal_amount)
                except ValueError:
                    print("Invalid input. Please enter a number.")
            elif choice == '3':
                print("\nThank you for using the ATM. Ejecting card.")
                break
            else:
```

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Code

```
            elif choice == '2':
                try:
                    withdrawal_amount = int(input("Enter amount to withdraw (multiples of 10): "))
                    atm.withdraw(withdrawal_amount)
                except ValueError:
                    print("Invalid input. Please enter a number.")
            elif choice == '3':
                print("\nThank you for using the ATM. Ejecting card.")
                break
            else:
                print("Invalid choice. Please enter 1, 2, or 3.")

if __name__ == "__main__":
    atm_session = ATM()
    atm_session.run_session()
```

```
Card inserted. Welcome, Sarmistha!
Enter your 4-digit PIN:  4321
🔑 Authentication successful.

--- ATM Menu ---
1. Check Balance
2. Withdraw Cash
3. Exit/Eject Card
Enter choice (1/2/3):  1
\ Your current balance is: 7000007000000.00

--- ATM Menu ---
1. Check Balance
2. Withdraw Cash
3. Exit/Eject Card
Enter choice (1/2/3):  2
Enter amount to withdraw (multiples of 10):  1000
 Dispensing $1000.00...
Please take your cash.
\ Your current balance is: 7000006999000.00

--- ATM Menu ---
1. Check Balance
2. Withdraw Cash
3. Exit/Eject Card
Enter choice (1/2/3):  3

Thank you for using the ATM. Ejecting card.
```

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